OPERATING SYSTEMS (CS304)

Page Replacement Algorithms

Learning Objectives

Understand virtual memory, demand paging, page faults, and major page replacement algorithms such as FIFO, LRU, and Optimal.

Introduction

Page replacement algorithms are used in virtual memory systems to decide which memory page should be replaced when a page fault occurs and no free frame is available.

Virtual Memory

Virtual memory allows processes to execute even when they are not completely loaded into main memory. It increases memory utilization and supports larger applications.

Demand Paging

Demand paging loads pages into memory only when they are required. This reduces memory usage and improves efficiency.

Page Fault

A page fault occurs when a process attempts to access a page that is not currently present in main memory.

Need for Page Replacement

• Limited physical memory
• Efficient memory utilization
• Support for multiple processes
• Reduced memory wastage

FIFO Page Replacement

First-In-First-Out (FIFO) replaces the page that has been in memory for the longest time.

Advantages of FIFO

Simple to implement and easy to understand.

Disadvantages of FIFO

May suffer from Belady's Anomaly and can replace frequently used pages.

LRU Page Replacement

Least Recently Used (LRU) replaces the page that has not been used for the longest period of time.

Advantages of LRU

Provides better performance than FIFO and approximates optimal behavior.

Disadvantages of LRU

Requires tracking page usage history, increasing overhead.

Optimal Page Replacement

Optimal algorithm replaces the page that will not be used for the longest time in the future. It provides the minimum possible number of page faults.

Numerical Example

Reference String: 7, 0, 1, 2, 0, 3, 0, 4

With 3 Frames: FIFO and LRU can be applied to calculate page faults and compare performance.

Belady's Anomaly

Belady's Anomaly is a situation where increasing the number of page frames results in an increased number of page faults. FIFO is susceptible to this anomaly.

Comparison of Algorithms

FIFO: Simple, but may suffer from Belady's Anomaly.
LRU: Better performance but higher overhead.
Optimal: Best performance but impractical in real systems.

Exam Important Questions

1. Define Page Fault.
2. Explain FIFO Page Replacement.
3. Explain LRU Page Replacement.
4. Explain Optimal Page Replacement.
5. What is Belady's Anomaly?
6. Solve numerical problems on page replacement algorithms.

Summary

Page replacement algorithms play a critical role in virtual memory management. FIFO, LRU, and Optimal algorithms differ in complexity and performance but aim to reduce page faults efficiently.